2 Digital literacy: Computers and learning

You will learn

- how computers are used to help people learn
- how computers are used to make discoveries
- how to use online research to help complete a project.



You can use computers to help you in any subject you study. The skills you learn in your computing classes do not just help you to learn while you are at school. They will be important throughout your life. When you leave school, computers will be important in your career.

Computers have been used to make important discoveries. You will learn about some of these discoveries in space exploration, medicine, and in understanding natural disasters such as earthquakes. Perhaps in the future you will contribute to a major discovery.

Later in the unit, you will work in small groups to create a short e-learning course. You will use online research to help you complete the project.



Unplugged

How do you use computers to help you do your schoolwork? Make a list. What software and hardware do you use?

Give examples of times you have used a computer to make work you are proud of.

Learning outcomes: Carry out an online research project; Explain how computers can help with learning and discovery

Did you know?

Scientists in a group of British universities created a 'robot scientist'. They claim it is the first computer to have discovered new scientific knowledge. The robot is called Adam. Adam uses **artificial intelligence** to carry out research. Artificial intelligence is a technique that programs computers to think and work like humans. Adam made a discovery about yeast. The discovery was simple. But it was important because it was made without much human help.





Talk about...

Computers can support your work in many subjects, not just computing. In which subjects do you use computers most?

Are there any subjects where you don't use computers at all? Why do you think that is?

artificial intelligence
assistive technology
computer models
e-learning
guidance computer
interactive video
multimedia
simulation
supercomputer
virtual reality
speech generator

2.1 Using computers for learning

In this lesson

You will learn:

- how computers can help you learn and discover information in all the subjects you study.

Spiral back

Computers are not just for computer studies. You can use computers to help you learn in any subject.

You have already learned some skills that will help you in your studies. You have learned how to:

- find information, to help you make the most of the web
- present information using word-processing, spreadsheet and presentation software.

In Student Book 5 you learned how to use a web browser to find information on the web. The skills you have gained in writing search strings should help you with other subjects like geography and history.

What do you already know?

Here is a reminder of some of the skills you have already learned.

You already know how to:

- **identify key words to use in a search**

  Think about the question you are trying to find answers to. Write your question down and underline the key words. This will help you identify what must be included in your web search.

- **use an age-appropriate web browser**

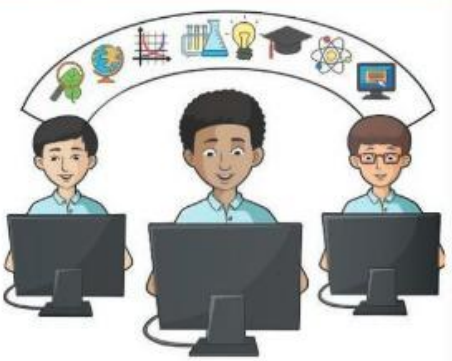
  Using a browser like Kiddle will show you information that is suitable for your age. It will filter out adverts.

- **use special search words and characters**

  You can use special characters in a web browser to make searching easier. For example, you can use the – (minus) sign to exclude terms from the search

- **check the reliability of information you find**

  In Student Book 5, you learned how important it is to check the sources of your information. Can you identify who wrote the information? Does the site publishing the information have a good reputation? Can you check key facts in the information on another site?



- **use bookmarks**

  Bookmarks make sure you can find useful sites again when you need them.

- **give credit for another person's work.**

  If you use content that belongs to another person, you should always indicate where the content comes from.

Now you have started to develop these skills, you can use the web to improve your work in all the subjects you study.

Activity

Choose two subjects that you study other than computing. For each subject, pick a topic that you are studying or have recently studied. Find one or two websites that provide useful information on these topics.

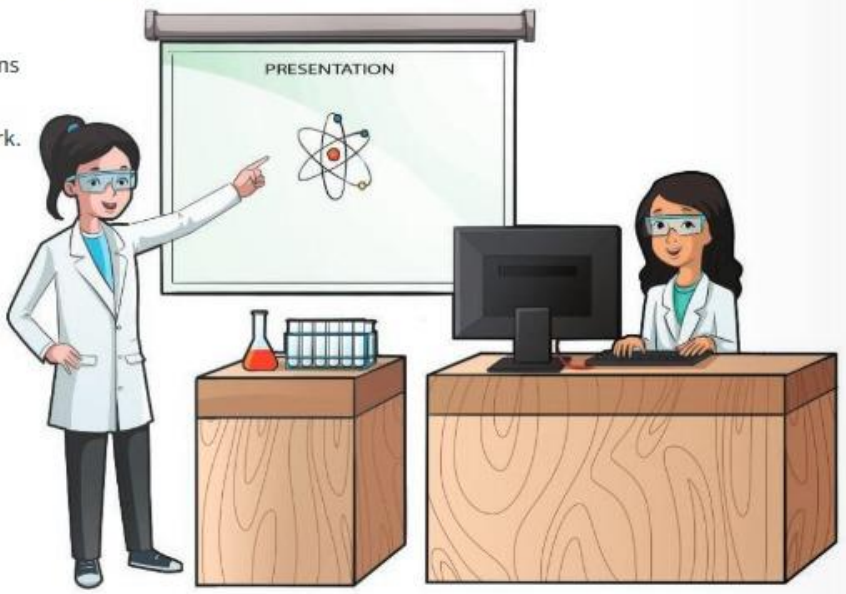
Share the sites with a small group of classmates. Did any of your group suggest other sites you will find useful?

Presenting your work

Think about the skills you have developed using application software. You have used a word processor, a spreadsheet and presentation software. You have also used graphics software to make images. All these skills are useful in any subject you are studying.

You can use application software to:

- write important course notes
- write assignments and homework
- create posters
- make presentations
- add images to illustrate your work.



Explore more

What are the advantages of using a word processor instead of writing your work by hand?

How much do your parents use word-processing apps in their work/at home rather than writing things out by hand? Discuss with them the advantages and disadvantages of using a word processor.

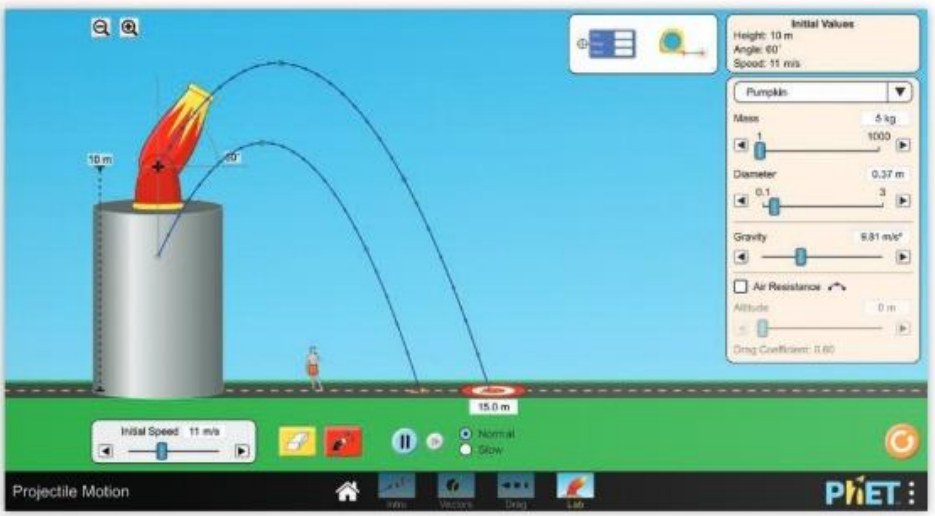
Other uses of technology for learning

Simulations

A computer **simulation** is a program that imitates something you can see or experience in real life. A simulation is like a game you play on your computer. You are in control of objects in a game. You can change how fast they move and the direction they move in.

You can change the way a simulation works. Simulations and games are interactive. This makes simulations a very useful tool for learning.

In science, simulations can be used for experiments. Chemistry experiments with dangerous chemicals can be simulated without students being at risk of injury. In geography, students use simulations to explore how glaciers are formed, or how greenhouse gas emissions affect global warming.



Here are some of the advantages of using simulations.

- They can be done anywhere.
- Experiments can be done many times.
- They speed up processes that take a long time.
- You don't have to set up complicated and expensive equipment.
- They can be used in situations where a real experiment is difficult to do.

Activity

Search the web to find a computer simulation. Use a search string like: 'free computer simulation science elementary school'.

Did you like using the simulation? Why or why not? Was it easy to use?

Virtual reality

Some schools have started to use **virtual reality** (VR) headsets for learning. You may be familiar with virtual reality from computer games. Wearing a VR headset gives you the feeling that you are in a different place. A virtual world is created around you. The world is interactive, like a simulation.

If you are learning about a place in geography or history, you read a description of the place. You can look at pictures or watch videos. With a VR headset, you can step inside the place. You can walk around and explore it.



Assistive technology

Assistive technology helps people with disabilities to learn.

Software can make on-screen text larger and clearer to help students with poor vision. Screen readers read out on-screen text for those with little or no vision. Braille converters translate on-screen text into a pattern of dots called braille. Braille is used as a way of reading by people who have no sight.



Some students have disabilities that make it difficult for them to speak. A **speech generator** turns text that you type into speech, using a simulated voice. The device saves phrases and words that you use often. This saves you from having to re-type everything.



Activity

A mouse can be difficult to use for some people with physical disabilities.

Search the web and find three devices that you can use instead of a mouse.

Extra challenge

Choose one of the devices from the activity.

Make an information sheet for the device you have chosen. Include a picture and write one or two paragraphs to describe how the device works.

Test

1. What is a computer simulation?
2. Give two examples of how assistive technology helps people with disabilities.
3. How can virtual reality be used to help people learn about geography?
4. How do simulations help people learn?

2.2 Using computers for discovery

In this lesson

You will learn:

- how computers can help make discoveries.

Space discovery

The Eagle has landed

On July 20, 1969 the Apollo Lunar Module called Eagle made the first landing on the moon. Neil Armstrong and Buzz Aldrin walked on the moon before returning safely to Earth.



Six Apollo missions to the moon returned safely to Earth. One of the reasons the Apollo missions were safe and successful was the Apollo **Guidance Computer** (AGC).

The AGC was built specially to guide spacecraft to the moon and back. It was the first computer used to guide flights. Today, all aircraft use guidance computers.

Compared with modern computers, the AGC was heavy, slow and difficult to use. It weighed as much as ten modern laptops. The computer chip in a car satellite navigation system is many times more powerful than the AGC.

Activity

There are programs that simulate the Apollo Guidance Computer. They show you how the computer looked and how to enter instructions.

Search the web for 'Apollo Guidance Computer Simulator'. What are the input and output devices it uses? How is it different from the computers you use?

Life on Mars?

Robots are used to explore planets. In January 2004, two identical robot explorers called Spirit and Opportunity landed on Mars. The explorers were supposed to work for 90 days. Opportunity worked for 15 years.

The robot explorers were designed to study the rocks on Mars in the same way that a geologist studies rocks on Earth.



The explorers had:

- two video cameras acting as eyes
- a flexible robot arm to examine rocks
- a hammer to break rocks open
- instruments to analyse rocks
- a microscope to look closely at rocks.

Spirit and Opportunity did not discover life on Mars. They did discover that ancient Mars had water and hot springs that could have supported simple microscopic life.

Activity

Look at the picture of the robot explorer. It seems to have a wing on each side. It doesn't fly.

What are the wings for?

Predicting natural disasters

Weather forecasting

Weather events like hurricanes can be devastating and life-threatening. Understanding how weather events form and move lets scientists called meteorologists predict where bad weather will occur. Early warnings about bad weather gives people time to prepare, or to evacuate an area.

There are two parts to predicting weather.

1. Satellites and radar record real weather conditions. Data about the weather is fed into a computer.
2. The weather data is processed using algorithms that predict how weather conditions will change over the coming hours and days.

This allows meteorologists to predict the course of hurricanes. They can warn people days before the hurricane arrives. The predictions are compared with what really happens. This is so meteorologists can discover more about the way storms work. They change their algorithms to help improve predictions in the future.



Very fast and powerful computers called **supercomputers** are used for this work. A supercomputer can process huge amounts of data very quickly.

Earthquakes

Another type of natural disaster is an earthquake. Parts of the earth move until rocks under the surface break. Huge forces cause the earth to shake violently. Buildings are damaged. Sometimes this causes them to collapse.

Scientists who study earthquakes are called seismologists. Seismologists use sensors to detect movement in the earth. They use satellite images of the earth. They feed this data into a computer model. Algorithms in the model help seismologists discover how earthquakes work.

Scientists now know that an earthquake is going to happen, but they don't know when. They hope that supercomputers will be able to predict earthquakes in the future. Predicting earthquakes will allow people to prepare for them. It will save many lives.



Activity

What preparations could you make if you knew in advance that a hurricane or earthquake was going to happen in your town? How could people be warned?

Medical discovery

DNA

Understanding DNA has led to many important medical discoveries in the 21st century. DNA research involves millions of calculations on large amounts of data. This has only been possible using computers.

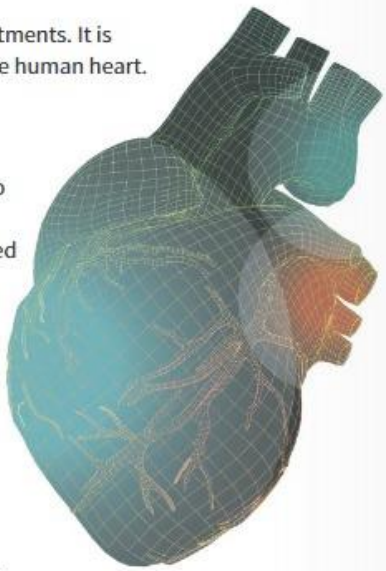
Scientists believe that DNA computers will be developed in the future. A microscopic computer made of DNA will be able to travel through the body and repair cells that are damaged by cancer or other diseases.



Computer models and medicine

Meteorologists and seismologists use **computer models** to make discoveries and to predict events. Other scientists use models too. Doctors and medical scientists use computer models. One use for models is to predict the effect that new treatments will have on patients.

Computer models are used to develop new drugs and other treatments. It is important to know if a new medicine will have a bad effect on the human heart. A computer model can help doctors to find out.



Animal and human testing

Drugs are usually tested on animals before permission is given to use them on humans. Before a new drug can be used, it must be tested on a small group of humans to check it is safe. This is called a clinical trial.

Many people think testing new medicines on animals is cruel. Clinical trials are usually safe but can be dangerous for the people involved. Doctors would like to develop more computer models so that they can test drugs using computers. Some medicines are already tested without using animals.

Activity

Why do you think it is important to use computers to understand and predict natural disasters? Give three reasons.

Extra challenge

Do you think testing medicines on animals is the right thing to do? Would you trust a medicine that has only been tested using a computer?

Outline the advantages and disadvantages of testing medicines using just a computer.

Test

1. Why does a meteorologist use computers?
2. Name one major space discovery that used computers.
3. What are computer models used for in medicine?
4. Name three pieces of equipment used by Mars explorer robots. Can you think of another piece of equipment that would be useful for researching the surface of a planet?

2.3 What is e-learning?

In this lesson

You will learn:

- how computers are used for e-learning.

What is e-learning?

An important use of computers in education is for **e-learning**. The 'e-' in e-learning stands for electronic. E-learning is like classroom learning in many ways, but it takes place over a local area network or the internet.



With e-learning, you can decide when and where you want to study. E-learning doesn't have a timetable, although there are usually set dates to submit your work. In the classroom, materials may be printed on paper or projected on the whiteboard. In e-learning, all the learning is done on a computer screen.

Who uses e-learning?

E-learning is often used in the workplace for training people. It is used to help people get new skills, for example, IT skills. It is also used when people need to know about new rules or ways of doing things, for example, if a government passes a new law that people in a particular job must know about.

E-learning is never the main type of learning in schools. It may be used by teachers to give you extra help. E-learning can also be used to give you extra challenges. Students can use e-learning at home to do extra study.

Multimedia and e-learning

In your classroom you have books to learn from. Your teacher gives you information. Sometimes, this information is given to the whole class using the board. At other times, you might work in small groups around a table. Your teacher may show videos or give a presentation using a projector.

There are many ways you can learn in the classroom. E-learning is no different. Many different types of media are used for learning. When different types of media, such as video and text, are used together, it is called **multimedia**.

Here is some of the media used in e-learning.

- **Video and interactive video**

  Video can be used for demonstrations, for example, how to use a piece of software. Or it can be used to show what an animal is like, while a commentary or words on-screen give facts about the animal.

  Sometimes you can decide how a video is played. For example, a video clip might be followed by a question. The next clip you see depends on what your answer to the question is. What you see depends on the action you take. This is called **interactive video**.

- **Presentations and audio**

  You have created presentations in earlier books. Some e-learning lessons are shown as a presentation. Presentations show information on a series of slides. An audio file is sometimes used to explain the content on each slide.

- **Text and images**

  Some e-learning material is made up of simple text with images, like the books in your classroom. Text and image content can be cheap and easy to make. It can be created quite quickly by teachers who do not have the IT skills to make videos, games and simulations. Books that you read on-screen are called e-books.

- **Games and simulations**

  You must complete learning challenges to progress in an e-learning game. Games can make learning fun.

  As we saw in Lesson 2.1, a simulation is a model of a real-life situation, for example, a scientific experiment. In a simulation of a chemistry experiment, you can safely see what happens if you increase the temperature or change the number or amount of chemicals.



Some students learn best if they watch a video. Others prefer to read, or to use a simulation. E-learning uses a mix of media types. That allows students to learn using the materials that suit them best.



Activity

How do you prefer to learn things? Do you prefer to read or watch a video? Do you think an educational game would help you learn?

How important is it for your teacher to explain something to you?

Tests and assignments

When you study at school, you do tests and exams. You do assignments and complete homework.

There are also tests and assignments in e-learning.

Tests are made up of short questions. For each question, you choose an answer from a list of options, or provide a single-word answer. This type of question can be marked automatically by a computer. This saves time and provides you with instant results.

Assignments are longer pieces of work that a trainer or teacher must mark. You send the assignment using the computer and get your mark back in the same way.

How is e-learning taught?

Technology is changing the way that people work. Trainers and teachers are still important in e-learning but their job is not the same as in classroom teaching.

In e-learning, teachers and trainers spend less time:

- teaching classes of students
- marking work
- working face to face with students.



This gives them more time to:

- help students get better results
- talk to students online and answer messages
- help individual students
- create e-learning media.

More learning is delivered on-screen, so a trainer or teacher does not spend as much time teaching in front of a full class. This gives them more time to spend helping individual students.

Advantages and disadvantages of e-learning



Advantages

Here are some of the advantages of e-learning.

- You can learn where and when you want to.
- If you live in a remote area, you can learn without having to travel long distances.
- You can use the type of media you prefer.
- Teachers have more time to help individual students.

Disadvantages

And here are a couple of disadvantages.

- Some people prefer to learn in a class with other people.
- It can be difficult to work without a strict timetable.

Digital citizen of the future

Technology is changing quickly. This means you will need to learn new skills throughout your working life. E-learning is one way for you to keep your skills and knowledge up to date.

Creating e-learning materials is a job that combines creative work with computing skills. Is it a career you would consider in the future?

Activity

Plan a lesson for a subject you are interested in.

- State the topic of the lesson.
- Outline the information you will include in your lesson.
- Write down one or two questions you could ask to see if the learner has understood the multimedia content.

Extra challenge

Do online research to find an example of multimedia content you could use for the lesson you planned.

Test

1. Explain what multimedia means in the context of e-learning.
2. Give three examples of media types that can be used for e-learning.
3. List three ways that a teacher's job will change if e-learning is used in your school.
4. Describe three ways that e-learning could help you to learn.

2.4 Starting your e-learning project

In this lesson

You will learn:

- how to plan your e-learning project.

Planning your e-learning project

For the rest of this unit, you will work in small groups to create a short e-learning course. There are some suggested topics below, but you can also choose your own topic.

Each member in the group will produce:

- a slide that covers one lesson
- a test question.

You will make the e-learning content using presentation software. The examples in this unit use Microsoft PowerPoint. If you use different presentation software, your teacher will tell you the differences between your software and PowerPoint.



First, plan how your team will work together. Here are some things you need to think about.

Choose a topic

It's up to your team to choose a topic. You could choose something you have studied in your computing course. Or choose a topic from another school subject. Here are some computing ideas.

- Network hardware (Unit 1 of this book)
- Computers and space discovery (see Lesson 2.2)
- Computer hardware (input, output and storage devices)
- Cybercrime (Unit 2, Student Book 7)
- Staying safe on the internet (Unit 2, Student Book 7).

Agree who will do what

When you have chosen a topic, decide how you will split it up into lessons. You need one lesson for each member of the group. For example, if you choose computer output devices, you could decide to divide your course like this:

- What is an output device?
- Computer screens
- Printers
- Speakers.

You need one heading for each member of your team.

Design your presentation

When you have completed your e-learning lesson, you will put it together with the lessons made by other members of your team. The completed presentation needs to look good. Each lesson needs to have the same style so that it looks like one person has written the whole course.

Here are some ways you can help ensure you have a consistent style.

- Agree a **theme** to use for your completed course. Your presentation software will have backgrounds you can choose from.
- Agree a **template** to use for your content. Your presentation software will have templates you can use. A layout like the one below will be suitable. There is room for a heading and an image with text next to it.

The themes that you can apply to your presentation are in the 'Design' tab. Choose one with a white background. It will make your text easier to read.

You can see what page templates are available if you click the 'Layout' button in the 'Slides' section in the 'Home' tab.

Your teacher will show you an example of a completed lesson.



Starting your project

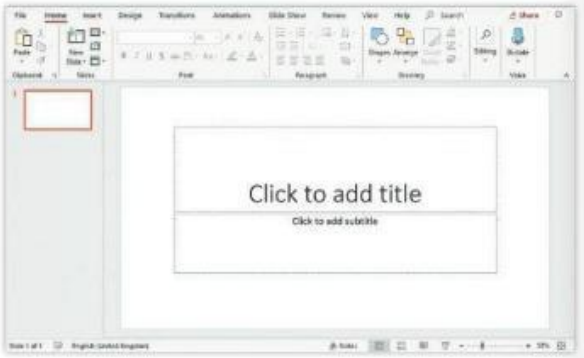
In Lesson 2.5, you will create your individual e-learning slides.

First, you will complete the title slide and contents page for your e-learning course.

Add a title slide

To add a title slide, open a new, blank presentation in your presentation software.

A blank title slide should be automatically inserted. If not, add a title slide.

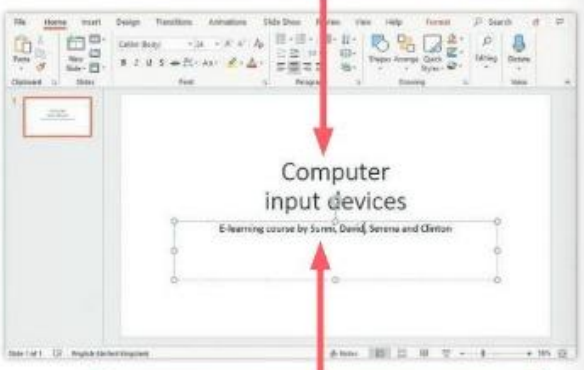


Add a title and credits

Now you are going to add a title and credits to your e-learning course.

Agree a suitable title for your presentation. It needs to describe what your e-learning course is about. It must not be too long.

1 Add a title to your presentation.



2 Add the names of your team members.

Choose a theme

Adding a theme to your presentation will make it look more professional. The theme you choose will appear on every slide, so try a few background designs before you make your final choice. Pick a theme that leaves plenty of space in the centre of your slides. A white background will make your text easier to read.

1 Select the 'Design' tab in your presentation software.

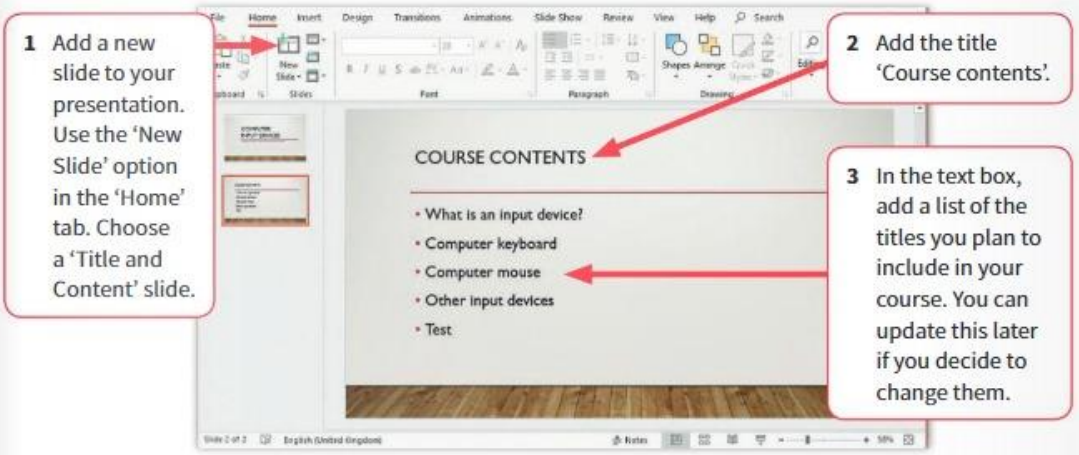
2 Browse the 'Themes' menu and select a theme.



Add a contents slide

Create a contents slide for your e-learning course. A contents page tells students what they will learn about when they study a course.

The contents slide should list the titles of the lessons your team members are going to create.



Save your work. You will use the presentation again in the next lesson.

When each member of your team has finished their slides, you will need to bring them all into a single presentation. Discuss how your team will do that.

Activity

Create the title and content slides for the e-learning course you have planned.



Extra challenge

Add an image to the title page. Choose an image that is relevant to your presentation. You can select an image using the 'Online Pictures' option in the 'Insert' tab.

Test

1. Why does your e-learning course need a title slide?
2. How do you add a new slide to a presentation?
3. Give reasons why you chose the theme for your presentation.
4. How will you carry out research for your e-learning lesson?

2.5 Create an e-learning slide

In this lesson

You will learn:

- use a presentation package to create a lesson.

In the last lesson, your team decided on a topic for the e-learning course. You each agreed to complete a slide for your group's e-learning course.

In this lesson you will create the lesson slides. Before you open your presentation software, do some planning and research for your lesson.

Activity

Write down the title of the e-learning course.

Underneath, write the title of your lesson.

Spiral back

In Student Book 5, you learned how to use key words to search for information on the web. Here is a reminder of the guidelines for using key words.

Make sure you understand the question you need to answer. Before you enter anything into the search engine:

- make a list of key words
- underline the most important key words
- make sure that the question you type in the search engine contains the important key words. Your question does not need to include punctuation or short joining words like 'is' and 'the'.

Research your content

Before you do any research on the web, write down what you already know about the lesson topic.



Activity

Make a list of short points that you already know about the lesson topic.

Add them under the lesson title you wrote down in the last activity.

Once you have written down what you know, it is time to do some research. There may be other facts that you want to include in the lesson.



Search the web

Before deciding what you will include in your lesson, search the web.

Look for:

- new facts. Your search may find facts you had forgotten to include in your list and some facts you weren't aware of.
- an image to illustrate your lesson. The image currently used in this lesson (a computer keyboard) isn't very helpful. In this case, it may have been more useful to include an image where the main parts of the keyboard are labelled.

Remember to check facts that you find on the web before you include them in your lesson. One way to check facts is to see if you can find the same fact on another website.

Remember that you will need to provide a credit in your presentation for the image that you choose. Make a note of the website you took the image from. You can use that information to credit the image.

Activity

Search the web for facts about your lesson topic. Choose two or three facts. Add them to the list of points you created in the earlier activity.

Search the web for an image that you can use in your lesson. Save the image.

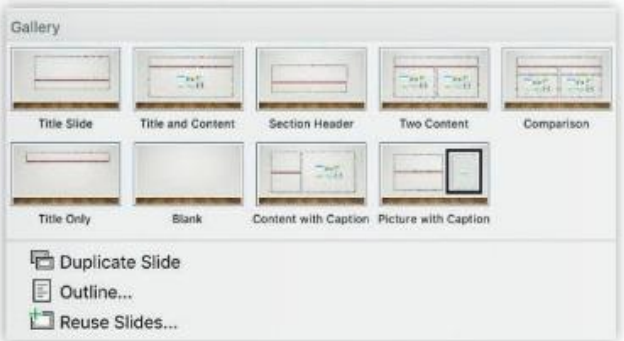
If you find more than one good image, save them all. You can decide which one to use later.

Create your slide

When you have all your lesson information, copy it into a slide. Remember to work as a team to make sure your slides all look the same.

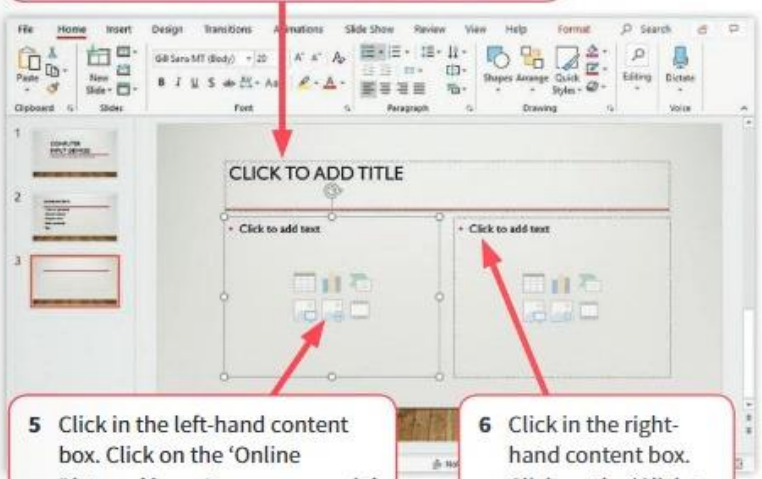
1 Open your presentation software.

2 Open a new slide. Select a template. The example in this lesson uses a template called 'Two Content'. The template has a space for a title. Below the title are two content boxes. You will add text into the right-hand content box and an image in the left-hand content box.



3 Add the theme you agreed with your team. You can then see how your content will look. Your page is ready for you to enter content.

4 Click in the title box. Type the title of your lesson. In our example, the title is 'Computer keyboard'.



5 Click in the left-hand content box. Click on the 'Online Pictures' icon. It opens a special browser that lets you search for pictures. Add an image.

6 Click in the right-hand content box. Click on the 'Click to add text' link. Add the text for the slide.

7 Save your file.

Your completed slide should look a bit like this.



Activity

Use the information you have gathered to create a slide with facts and an image. Use the layout shown in this example.

Work with the other students in your group to make a presentation which includes all the slides you made.

Extra challenge

Look through the whole presentation. Say what you like best about the other slides in the presentation. Finally, think if there are any ways that the presentation could be improved. If you have time, make those changes.

Test

1. Why is it important to identify key words before you do a web search?
2. What is a template used for in presentation software?
3. Why do you need to include a credit for images or content you use from websites?
4. Why is it important to check facts you find on the web before using them?

2.6 Add a test question

In this lesson

You will learn:

- how to add a test to your e-learning lesson.

Making a test

In Lesson 2.3 you learned that e-learning courses usually contain tests. This is so that students can check that they have understood the course content. Most e-learning tests are marked automatically by the computer. This means that you know straight away if you have given a correct answer. It also means that the teacher or trainer doesn't have to spend time marking the test.

Multiple-choice tests

The test you are going to write is called a multiple-choice test. Each multiple-choice question has more than one possible answer (usually four). Only one answer is correct. Here is an example.



The answer is B, Paris.

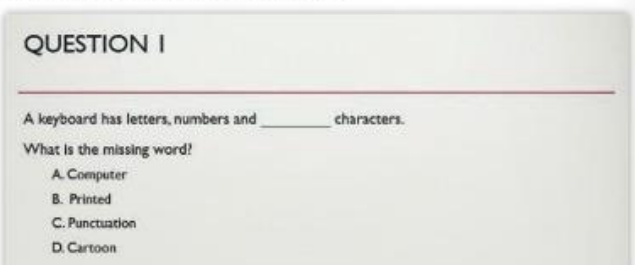
There is usually only one correct answer to a multiple-choice question. Simple questions like this one are easy for a computer to mark. If the answer is B, it is correct. Any other answer is incorrect.

Making an example question

Now you will make a test to find out whether students have understood the information in your lesson.

Work as a group to make a one-question test. It can be about any topic in your e-learning presentation. A student must be able to answer the question from information in the presentation.

The example lesson was about the computer keyboard. Here is an example test question. In this case, the question tests information from the third bullet point in the lesson.


QUESTION 1

A keyboard has letters, numbers and _____ characters.

What is the missing word?

A. Computer
B. Printed
C. Punctuation
D. Cartoon

Activity

Add a new slide. Select a 'Title and Content' slide.

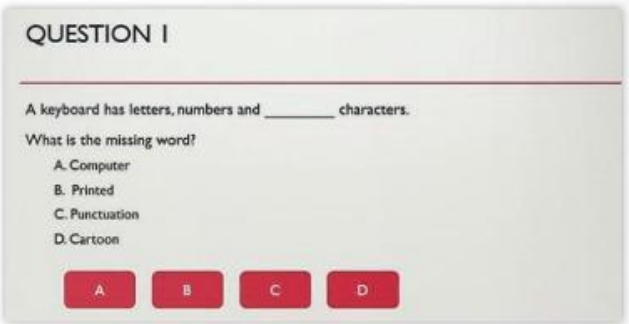
Type 'Question 1'. This is the title for your slide.

Type 'A keyboard has letters, numbers and _____ characters' followed by 'What is the missing word?'

Add four answers, like in the example: A. Computer, B. Printed, C. Punctuation and D. Cartoon.

Adding answer buttons

Once you have written your question, you must provide a way for students to give their answer. You will add buttons for the student to click.



Activity

Go to the 'Insert' tab and click on 'Shapes'.

Select a rectangle from the 'Shapes' menu. Draw a rectangle below your question.

Copy the first rectangle and paste it three times. That way you will get four buttons that are the same size. Insert the letters A, B, C and D on the boxes.

Making the buttons work

You now have a question slide with buttons to allow a student to give their answer.

Now you will add actions to each button to make them work.

The slides contain emoticons (icons showing facial expressions). If you want to add emoticons to your slides, click 'Icons' in the 'Insert' tab. The emoticons are under 'Faces'.



Activity

First, add three slides to your presentation. Add them in this order:

1. A slide with the heading 'That's right!' and a button that has the label 'Next question'.
2. A slide with the heading 'Wrong answer' and two buttons labelled 'Try again' and 'Next question'.
3. A slide with the heading 'Question 2'. This page doesn't need any other content.

Adding links

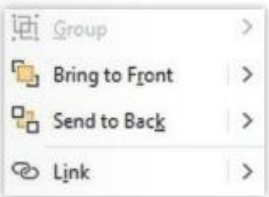
Now you are ready to add links that will tell your students whether they have the right answer.

Go back to the Question 1 slide you made earlier.

Activity

Right-click on the button labelled 'A'. A box should appear around the button. A menu will appear on-screen.

Click 'Link'. You will find this about half-way down the menu.

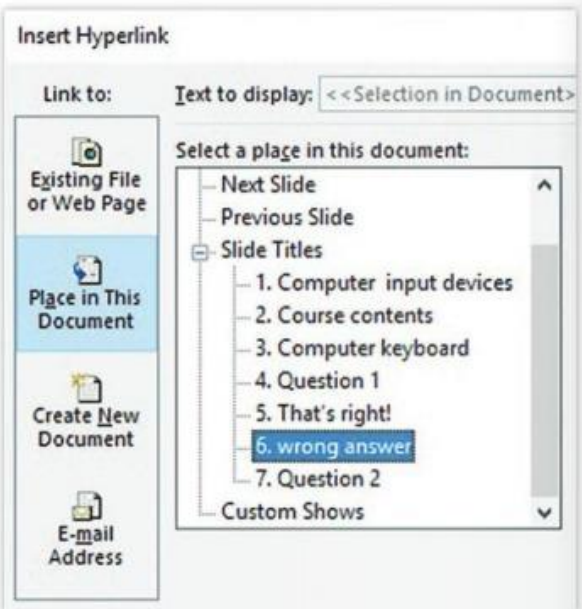


A box will open on your screen. There is a button on the left that says 'Place in This Document'. Click that button. A list of the slides you have made appears to the right of the button.

Choose the slide that you want button A to link to. Answer A is the wrong answer, so you should pick the Wrong answer slide.

Clicking button A on the Question 1 slide should now take you to the Wrong answer slide. Test it works by running your presentation. (Click the 'From Beginning' link in the 'Slide Show' tab.)

Once the link works, add links to the other buttons in your question slide. Here are the links you need to make.



Slide title	Button	Link to slide
Question 1	C and D	Wrong answer
Question 1	B	That's right!
That's right!	'Next question'	Question 2
Wrong answer	'Next question'	Question 2
Wrong answer	'Try again'	Question 1

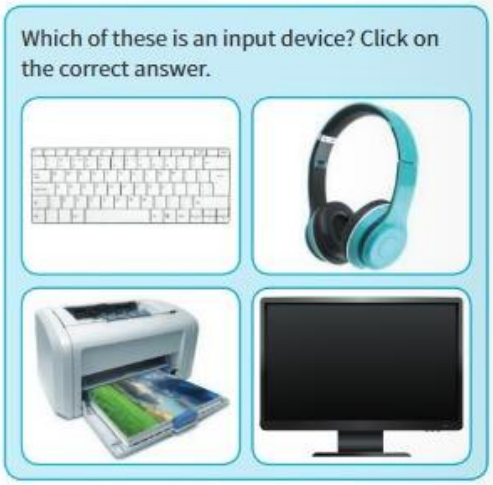
Other question types

You have created a multiple-choice question and answers in this lesson.

You could use other types of questions in your test. Here are a couple of ideas.

- True/False. You could give a statement such as: 'A printer is an input device.' Then ask the student to say whether the statement is True or False. This is a type of multiple-choice question but with only two options.
- Use pictures instead of words as possible answers. The picture on the right shows an example.

A picture question doesn't need buttons. You can add links to the pictures. Use the same method that you used for the A, B, C and D buttons.



Activity

This lesson showed you how to make a test question based on the computer keyboard example. Working as a group, make one test question that matches your e-learning lesson.

Extra challenge

Add more questions to the end of the presentation. You can do this working as a group, or each person can make their own slides and then copy them all into the group file.

Test

1. How do you run a slide show? What happens on your screen?
2. How many correct answers can you have in a multiple-choice test?
3. Explain how a link works in a presentation slide. Give an example of how you use a link.
4. Outline three types of test you could use. Why would you use different types of test?

Check what you know

You have learned

- how computers are used to help people learn
- how computers are used to make discoveries
- how to use online research to help complete a project.

Try the test and activities. They will help you to see how much you understand.

Test

1 Outline two ways that you have used computers to learn and find things out in subjects like science or geography.

2 Identify two types of natural disaster that can be researched using computers.

3 Describe how computers have been used in space exploration.

4 Give two advantages of using simulations for learning.

5 Describe how assistive technology helps people with disabilities to learn.

6 Describe how technology has been used to discover how natural events like hurricanes work. How have these discoveries benefited people?



Activities

In Lessons 2.4 to 2.6 you worked with a group to create a short e-learning course. Think back to that task and complete the following activities.

1. Give at least two examples of content or facts you found online when doing research for your e-learning lesson.
2. List the content items you used in your lesson. Where did you find the information? Give reasons why you chose that content. List one item of content you found that you decided not to use in your lesson. Why did you reject it?
3. Describe some of the presentation features you used to make the e-learning effective.



Self-evaluation

- I answered test questions 1 and 2.
- I completed activity 1.
- I answered test questions 1–4.
- I completed activities 1 and 2.
- I answered all the test questions.
- I completed all the activities.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?